Robot Vision: Mock Exam 1 (Solutions)

Chapters 2 to 12 · 90 minutes · no documents · 40 points

Part A: Multiple choice

22 points, about 35 min

A1 What does a standard neural-network unit compute? (ch. 2)

(a) The product of its inputs, compared with a threshold.
(b) The maximum of its inputs.
(c) The average of its inputs, without weights.
(d) ✓ **A weighted sum of its inputs plus a bias, passed through an activation function.**

$u = \sum_i w_i x_i + b$, $z = f(u)$.

A2 Why is multi-label classification implemented with independent sigmoids rather than a softmax? (ch. 2)

(a) Because sigmoid outputs always sum to one.
(b) Because the softmax is too slow to compute.
(c) ✓ **Because several labels can be true for the same input at the same time.**
(d) Because the labels are ordered.

Softmax would force the labels to share a total probability of 1.

A3 The training error keeps decreasing but the validation error starts to increase. What does this indicate? (ch. 3)

(a) ✓ **Overfitting.**
(b) Underfitting.
(c) That the test set was used for training.
(d) A learning rate that is too small.

The model fits the training set at the expense of new data.

A4 Which statement about weight decay is correct? (ch. 3)

(a) AdamW removes weight decay.
(b) ✓ **With Adam, weight decay is not equivalent to L2 regularization, so AdamW applies it separately.**
(c) Adam ignores weight decay.
(d) Weight decay and L2 are equivalent for every optimizer.

With plain SGD they are equivalent.

A5 What is a key difference between traditional recognition methods and CNNs? (ch. 4)

(a) CNNs do not use any classifier.
(b) CNNs use hand-designed features such as SIFT.
(c) Traditional methods learn their features end to end.
(d) ✓ **Traditional methods use hand-designed features and a separate classifier; CNNs learn the features from data.**

Feature learning is the big change.

A6 A 7×7 input goes through a 3×3 convolution with stride 2 and no padding. What is the output size? (ch. 4)

(a) 5×5
(b) ✓ 3×3
(c) 4×4
(d) 2×2

$\lfloor (7-3)/2 \rfloor + 1 = 3$.

A7 Why can a CNN with global average pooling accept images of different sizes? (ch. 5)

(a) It resizes every image to 224×224.
(b) It uses one fully connected layer per image size.
(c) It removes all convolutions.
(d) ✓ **It produces one value per channel, so the vector length does not depend on the spatial size.**

GAP replaces the size-dependent FC input.

A8 What does SSD predict for each default box? (ch. 5)

(a) Only a class score.
(b) ✓ **Offsets that adjust the box, and a score for each class.**
(c) A full segmentation mask.
(d) The depth of the object.

Then NMS removes duplicates.

A9 Why are the first-layer filters of a CNN easy to interpret? (ch. 6)

(a) They are always 1×1.
(b) They are designed by hand.
(c) They have more weights than deeper filters.
(d) ✓ **They act directly on the pixels, while deeper filters act on feature maps.**

They look like edges and color blobs.

A10 What does a class saliency map show? (ch. 6)

(a) The weights of the first filter.
(b) The activation of the last layer.
(c) The pixels removed by pooling.
(d) ✓ **The absolute value of the gradient of the class score with respect to each pixel.**

How much the score reacts to each pixel. Noisy.

A11 What must be done when applying a geometric augmentation to detection training images? (ch. 7)

(a) Only flip the class labels.
(b) Remove the boxes.
(c) Nothing, the boxes stay valid.
(d) ✓ **Transform the bounding boxes in the same way as the image.**

The target must stay valid.

A12 What is the difference between semi-supervised and weakly supervised learning? (ch. 7)

(a) Both use only labeled data.
(b) ✓ **Semi-supervised combines labeled and unlabeled data; weakly supervised uses incomplete or coarse labels.**
(c) Weakly supervised uses no data at all.
(d) Semi-supervised uses coarse labels; weakly supervised uses unlabeled data.

E.g. one label per bag.

A13 How can face identification work when the gallery identities were not in the training set? (ch. 8)

(a) It cannot work for unseen identities.
(b) By retraining the network for each person.
(c) ✓ **By comparing the query feature with the gallery features in a learned metric space.**
(d) By adding a softmax unit for each new person.

Metric learning: distance means identity.

A14 What happens when the softmax temperature T is increased? (ch. 8)

(a) All probabilities become zero.
(b) The predicted class changes.
(c) The distribution becomes sharper.

(d) ✓ **The distribution becomes flatter, and the predicted class does not change.**

Used for calibration.

A15 What is the fundamental difference between sequence data and set data? (ch. 9)

(a) Sequences have no order.
(b) Sets can only contain numbers.
(c) ✓ **Order matters in a sequence but is arbitrary in a set.**
(d) Sets are always longer than sequences.

Sets need invariant or equivariant networks.

A16 How does PointNet get a permutation-invariant global feature? (ch. 9)

(a) Each point gets its own MLP.
(b) ✓ **A shared MLP is applied to each point, then max pooling (a symmetric function) combines them.**
(c) An RNN reads the points in order.
(d) The points are sorted first.

Concatenate global and point features for segmentation.

A17 What is the key characteristic of an RNN? (ch. 10)

(a) It has no hidden state.
(b) Each step has its own weights.
(c) It processes all time steps in parallel with attention.
(d) ✓ **It updates its hidden state from the current input and the previous hidden state.**

$h_t = f(Wx_t + Uh_{t-1})$.

A18 How does ViT turn an image into tokens? (ch. 10)

(a) It applies a full CNN and keeps one vector.
(b) It uses overlapping patches without positions.
(c) ✓ **It cuts the image into non-overlapping patches, embeds them linearly, adds position embeddings and a class token.**
(d) It uses each pixel as a token.

224^2 image, 16×16 patches: 196 tokens + class token.

A19 Why is attention useful in visual question answering? (ch. 11)

(a) It gives the same weights for every question.
(b) It replaces the image encoder.
(c) ✓ **The relevant part of the image depends on the question, and attention weights the regions accordingly.**
(d) It makes the image smaller.

"What animal?" vs "Is it cloudy?".

A20 What happens to the model parameters during in-context learning? (ch. 11)

(a) ✓ **They are not updated; the examples are only given in the prompt.**
(b) Only the last layer is updated.
(c) They are reset to random values.
(d) They are fine-tuned on the examples.

Instruction tuning, by contrast, updates them.

A21 What does a generative model of images try to learn? (ch. 12)

(a) The labels of the training set.
(b) A mapping from images to class labels.
(c) ✓ **The distribution $p(x)$ of the images, so that new samples can be drawn from it.**
(d) The weights of a classifier.

Unsupervised: only x is given.

A22 How does a trained VAE generate a new image? (ch. 12)

(a) Encode a training image and decode it.
(b) Ask a discriminator to choose.
(c) Run the encoder on random noise.
(d) **✓ Sample z from the prior $\mathcal{N}(0, I)$ and decode it; the encoder is not needed.**

The KL term keeps codes close to the prior.

Part B: Short answers

12 points, 3 per question, about 35 min

B1 Why do residual connections make very deep networks easier to train? (ch. 3)

A block computes $z_{l+1} = z_l + F(z_l)$. The identity path lets the gradient flow back directly, so it does not vanish through many layers, and a useless block can simply learn $F = 0$.

B2 Name the two properties that distinguish a convolutional layer from a fully connected layer, and give one consequence. (ch. 4)

Sparse (local) connections and weight sharing. Consequences: far fewer parameters, and translation equivariance (the same detector is used everywhere).

B3 In FixMatch, why is the pseudo-label taken from the weakly augmented view rather than the strongly augmented one? (ch. 7)

The weak view is easier to predict, so a confident prediction on it is more reliable. It then serves as the target for the strong view.

B4 Explain how CLIP can classify an image without training on the target classes (zero-shot). (ch. 11)

Write each class as a text such as “a photo of a dog”, encode the texts and the image, and choose the class whose text embedding is most similar to the image embedding. CLIP learned this shared space contrastively from image-text pairs.

Part C: Calculations

6 points, 2 per question, about 15 min

C1 A 9×9 input goes through a 3×3 convolution with stride 2 and padding 1. What is the output size? (ch. 4)

$\lfloor (9 + 2 - 3)/2 \rfloor + 1 = 4 + 1 = \mathbf{5}$, so 5×5.

C2 How many parameters (weights and biases) does a 3×3 convolution from 16 to 32 channels have? (ch. 4)

$3 \cdot 3 \cdot 16 \cdot 32 + 32 = 4608 + 32 = \mathbf{4640}$.

C3 Five classifiers predict A, A, B, A, C for an unlabeled image. What is the variation ratio? (ch. 7)

Mode A with 3 votes: $1 - 3/5 = \mathbf{0.4}$.